



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

– Hira Kumar

OPERATING SYSTEM (OS)



❖ Disk Management	2
Goal/Feature/Function of Disk Management	2
Accessing Disk Management on Your Computer	3
❖ Disk / Disk Structure	5
❖ Disk Access Time	7
❖ Disk Scheduling	9
❖ Disk Scheduling Algorithm	9
FCFS (First come first serve)	9
SSTF(Shortest Seek Time First)	11
SCAN	13

C-SCAN		15
LOOK		16
C-LOOK		18

Q1. What is Disk Scheduling ? Discuss disk scheduling algorithms :
SCAN, CSCAN, FCFS, CLOOK. (2021 , 2022,2024)

=====

❖ **Disk Management**

- Disk Management is a core function of an OS that involves organizing,controlling, and maintaining the computer's storage devices (hard disks, SSDs, USB drives etc), It ensures efficient data storage, retrieval and organization.

➤ **Goal/Feature/Function of Disk Management**

- 1. Data Organization & Structure** → This is achieved by dividing physical storage into distinct, manageable partitions or volumes. By separating the operating system from user files, applications, and media, this organization enhances performance, simplifies file navigation, and creates a clear structure. It prevents the system drive from becoming cluttered, which can slow down critical processes. A well-organized disk is the foundational blueprint for a stable and responsive computing environment.
- 2. Storage Optimization & Space Management**
→ Disks often contain unusable "unallocated" space or partitions that are poorly sized for current needs. Disk management tools allow administrators to dynamically extend volumes into free space, shrink underutilized ones, and merge fragmented areas. This proactive space reclamation ensures that every gigabyte of

the physical drive is actively usable, delaying the need for costly hardware upgrades and preventing scenarios where one partition is full while another has ample free space.

3. Data Protection & Recovery Planning

→ Disk management facilitates this by creating dedicated partitions for system backups or by implementing software RAID configurations like mirroring (RAID 1). Isolating the OS on its own partition means a system failure or malware infection can often be addressed by wiping and reinstalling just that volume, leaving personal data on a separate partition untouched. This logical separation is a first line of defense, making disaster recovery faster and more reliable.

4. Performance Enhancement & hardware

Alignment → This involves aligning the storage architecture with the hardware's capabilities. For instance, using a modern GPT partition scheme for large SSDs, or strategically placing high-performance applications on the fastest part of a disk. For traditional hard drives, management allows for defragmentation within partitions. Properly configuring partitions also helps manage write cycles on SSDs. The goal is to tailor the logical layout to extract optimal speed and endurance from the physical hardware.

5. System Flexibility & Future Scalability → A single physical disk can host multiple operating systems (like Windows and Linux) by creating separate bootable partitions. Disk management also allows for dynamic resizing and reconfiguration without complete reformatting, accommodating changing needs. This forward-looking planning ensures the storage system is not a rigid constraint but an adaptable resource that can evolve with user requirements, software updates, and new technological demands.

❖ Accessing Disk Management on Your Computer

(अपने कंप्यूटर में disk management कैसे open/access करें)

✓ **Method 1** → Press Win+R → type diskmgmt.msc → press enter

✓ **Method2** → Control panel → System and security → Administrative tools → Computer management (यहाँ पर direct आने के लिए This pc → Right click → manage) → Disk management.

✓ **Method3** → Search hard disk partitions and select open.

- Volume → Volume means a logical partition of storage (e.g. – C drive, D drive, E: etc...)

- Layout → Shows how data is organized on disk. (disk पर डाटा किस तरीके से रखा गया है)

e.g. – Simple means single disk

Spanned means Multiple disks

Striped means performance

Mirrored means backup

RAID-5

- Type → Shows disk type – Basic disk (normal), Dynamic disk (advanced features like RAID)

- File system → Defines how files are stored and managed (e.g.- NTFS, FAT32, exFAT,...)

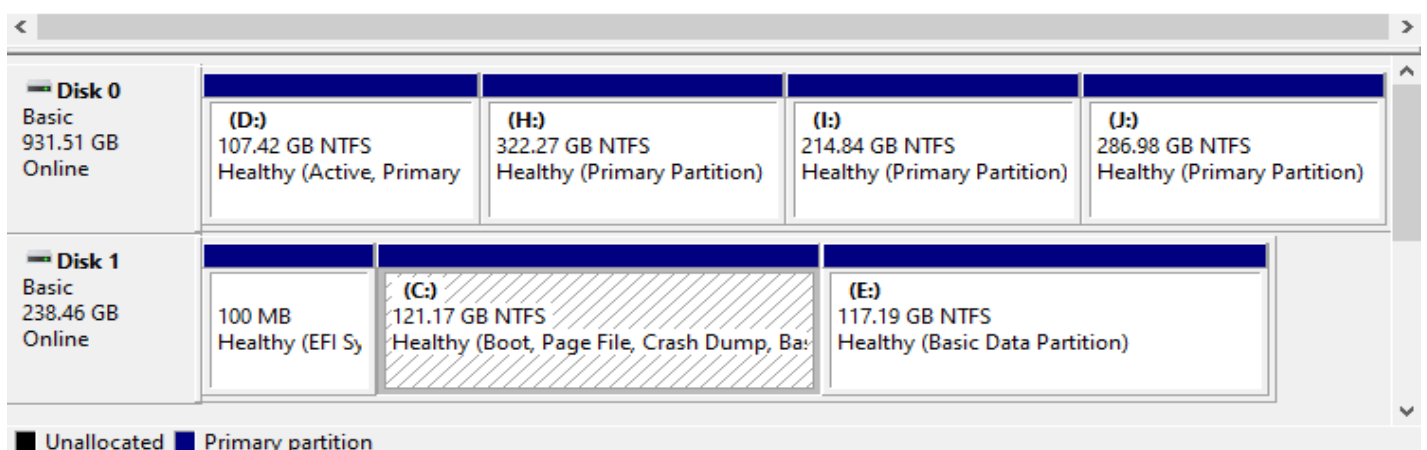
- Status → Shows current condition of volume (e.g. Healthy, Failed, Formatting, Unknow)

- Capacity → Total size of the volume

Disk Management

File Action View Help

Volume	Layout	Type	File System	Status	Capacity	Free Spa...	% Free
(C:)	Simple	Basic	NTFS	Healthy (Boot, Page F...	121.17 GB	34.38 GB	28 %
(D:)	Simple	Basic	NTFS	Healthy (Active, Prim...	107.42 GB	45.23 GB	42 %
(E:)	Simple	Basic	NTFS	Healthy (Basic Data P...	117.19 GB	14.73 GB	13 %
(H:)	Simple	Basic	NTFS	Healthy (Primary Parti...	322.27 GB	122.41 GB	38 %
(I:)	Simple	Basic	NTFS	Healthy (Primary Parti...	214.84 GB	47.45 GB	22 %
(J:)	Simple	Basic	NTFS	Healthy (Primary Parti...	286.98 GB	148.56 GB	52 %
(Disk 1 partition 1)	Simple	Basic		Healthy (EFI System P...	100 MB	100 MB	100 %



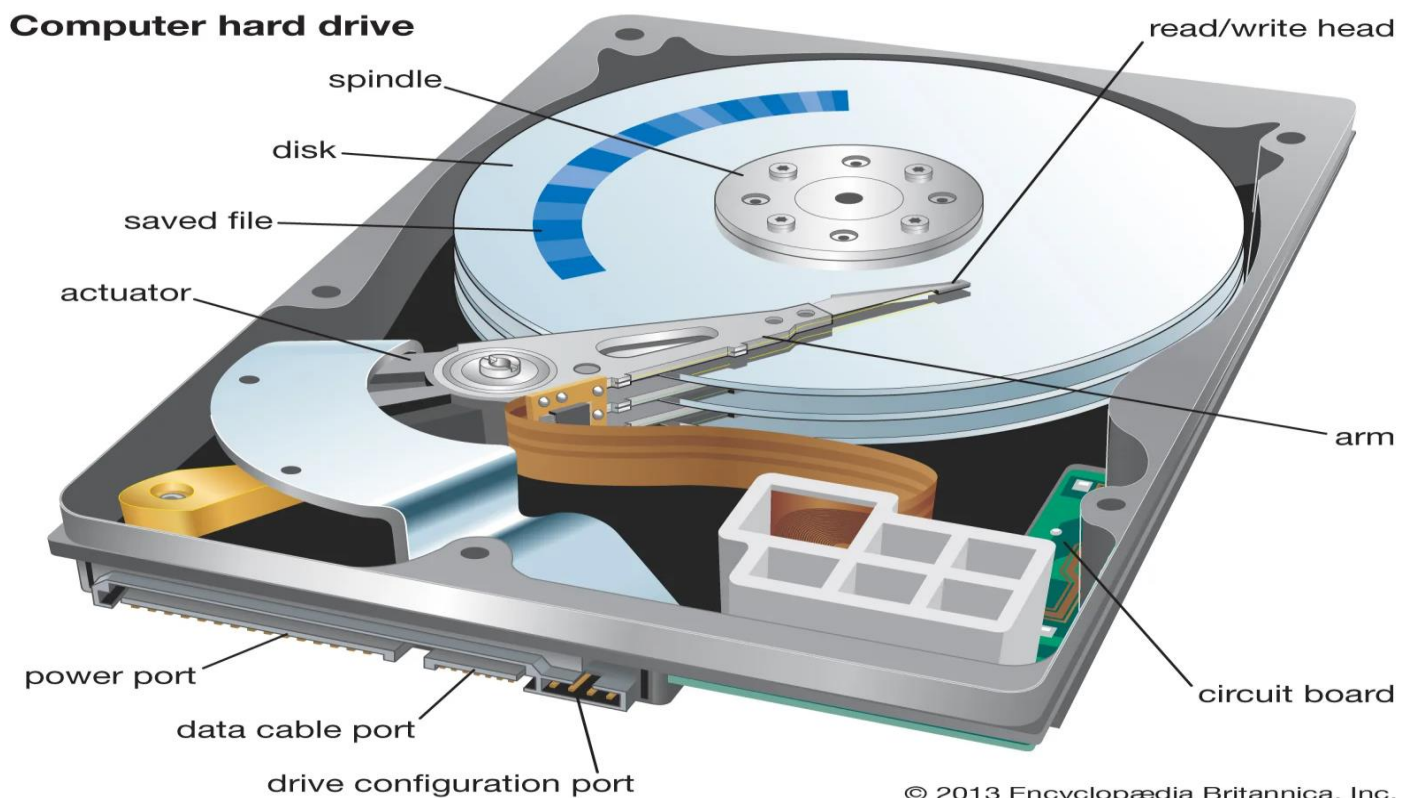
- Shrink Space → Shrink space means reducing the size of an existing partition and creating unallocated space from it.

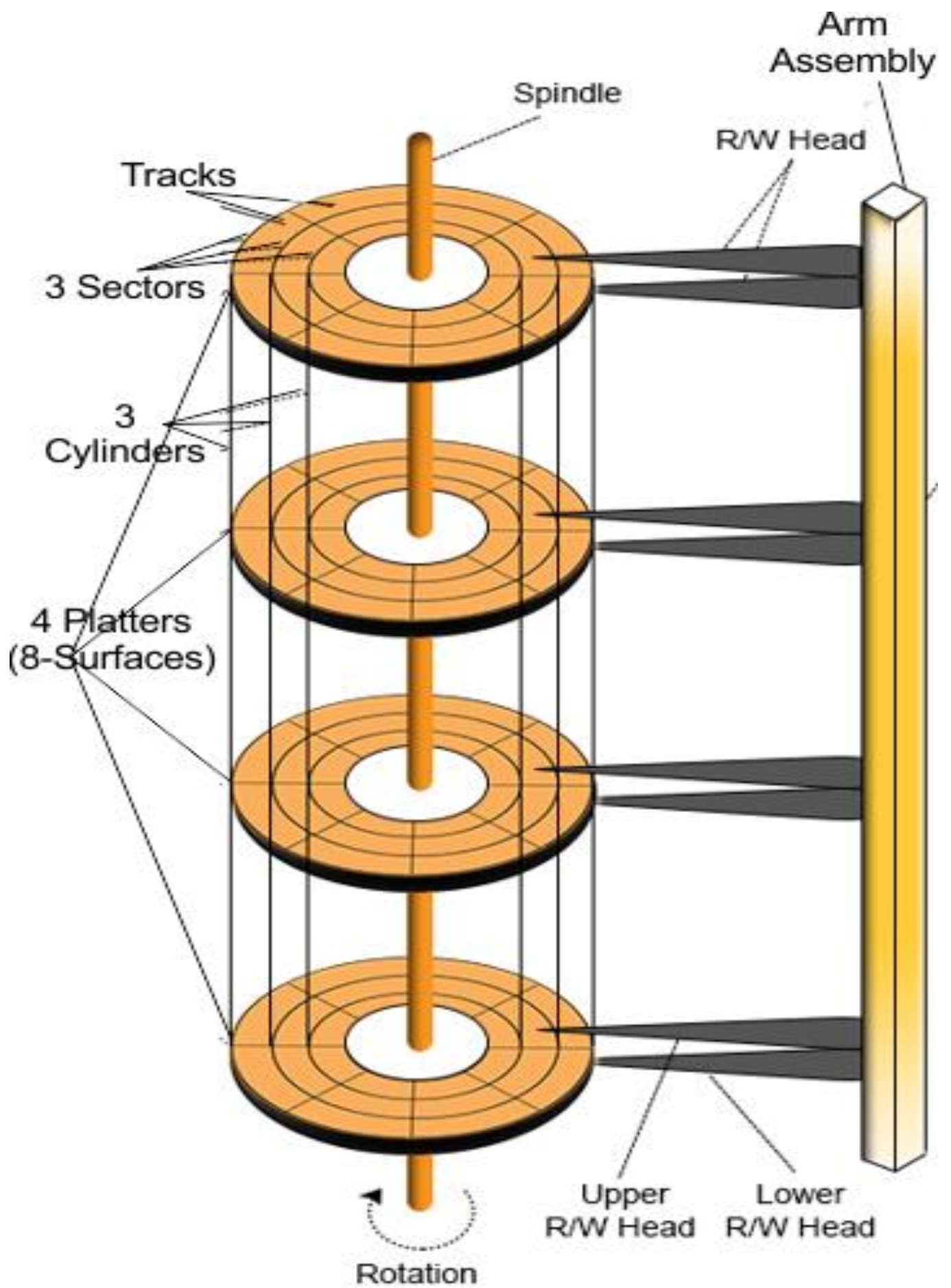
Example : If C drive = 100 GB and you shrink 20 GB then

C drive becomes 80 GB and 20 GB becomes unallocated space.

❖ Disk / Disk Structure

- A process needs two type of time, CPU time and IO time. For I/O, it requests the Operating system to access the disk.
- Each Disk has a flat circular shape like a CD or DVD.
- Generally Diameter range from 1.8 to 5.25 inches.
- Surface of disk is divided into circular track, tracks are further divided into sectors.



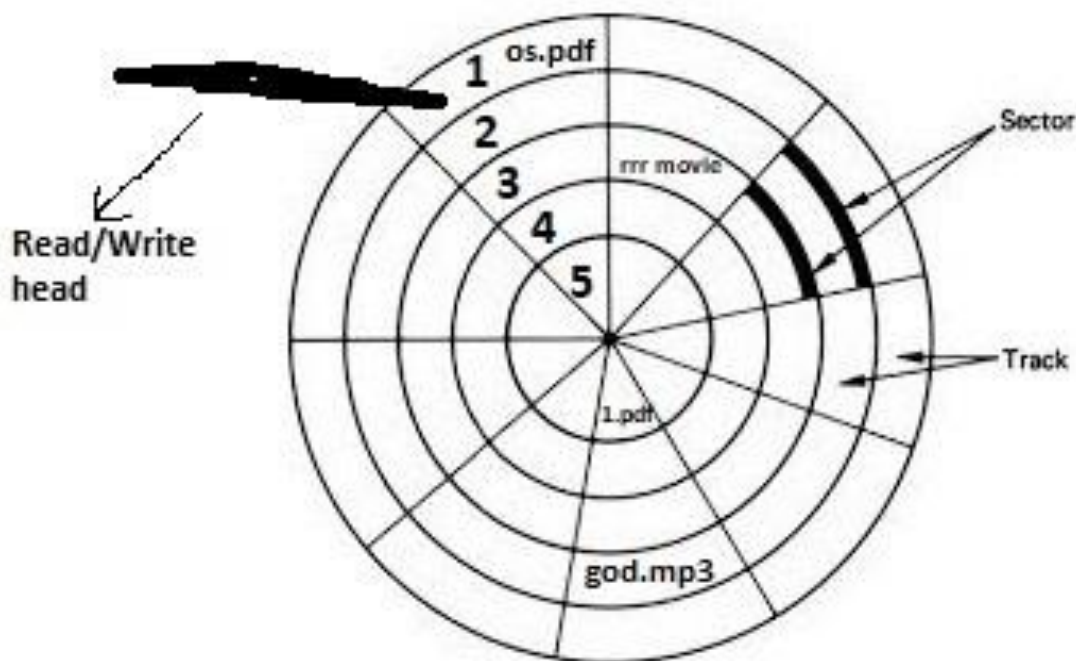


❖ Disk Access Time

- **Seek Time:** Time required to move the read/write head on the desired track. (जिस ट्रैक पर data रखा हुआ है, R/W head को उस ट्रैक पर जाने में लगा समय को हम seek time कहते हैं)

It is two type –

- ✓ **Best Case :-** In this case, the data remains/found the same place where the r/w head is available.
- ✓ **Worst Case :-** In this case, the data should be found away from the r/w head. OR data found last track where r/w head available 1st track.
- ✓ **Average Case :-** In this case, data found nearest track.



Example :-

यदि user os.pdf सर्च करता है तो उस condition में ये best case हुआ ।

यदि user 1.pdf सर्च करता हैं तो उस condition में ये worst case हुआ |

यदि rrr movie search करता हैं तो उस case में average case होगा|

➤ **Rotational Time**

Time taken for one full rotation (360°)

It is measure in RPM (Round per minute) e.g. = 7200 RPM

(disk platter को 1 पूरा चक्कर लगाने में लगा समय)

➤ **Rotational Latency / Latency Time**

Time delay b/w a request for data and the return of the data.

(head जिस sector को पढ़ना चाहता हैं, उसके हेड के नीचे आने तक का wait time)

➤ **Transfer time**

Time to actually read/write the data once the sector is located.

(sector head के नीचे आ जाने के बाद actual data read/write करने का समय)

➤ **Data transfer time**

Disk से memory तक भेजने में लगने वाला कम समय

➤ **Access time**

- Access time in a HDD is the total time it takes for the drive to locate and read/write data.
- Access time = Seek time+ rotational latency + data transfer time

➤ **Number of track =** यदि किसी HDD में 200 ट्रैक रखता हैं, तो उसमें कुल ट्रैक की संख्या $200-1=199$ होगा, क्योंकि यहाँ ट्रैक प्रारंभ 0 से होता हैं, मतलब कुल संख्या (0-199)

❖ ***Disk Scheduling***

- Service provide the disk I/O requests.
- Disk Scheduling is an operating system technique used to manage and decide the order in which disk I/O requests are serviced.
- Since multiple processes may request access to the disk at the same time, disk scheduling helps the OS choose which performance is improved.
- If desired disk drive or controller is available, request is served immediately.
- If busy, new request for service will be placed in the queue of pending requests. When one request is completed, the OS has to choose which pending request to service next.

❖ **Why disk scheduling**

Use hardware efficiently

Reduce disk access time by minimizing the movement of the disk head and improving system performance.

❖ **Disk Scheduling Algorithm**

In case of multiple I/O request disk scheduling algorithm must decide which request must be executed first.

✓ Many types of Disk scheduling algorithm

1. FCFS(First Come First Serve)
2. SSTF(Shortest Seek Time First)
3. SCAN
4. LOOK
5. C-SCAN(Circular Scan)
6. C-LOOK(Circular Look)

1. FCFS (First come first serve)

- The simplest & fastest form of disk scheduling.
- I/O requests are serviced based on their arrival.

✓ **Advantage:**

- Easy to understand
- Easy to implement
- It can be used with less load of process

✓ **Disadvantage**

- Required more Seek time and Waiting Time.

Q1. Consider the following disk request sequence for a disk with 90 tracks.

12, 34, 52, 14, 25, 68, 39.

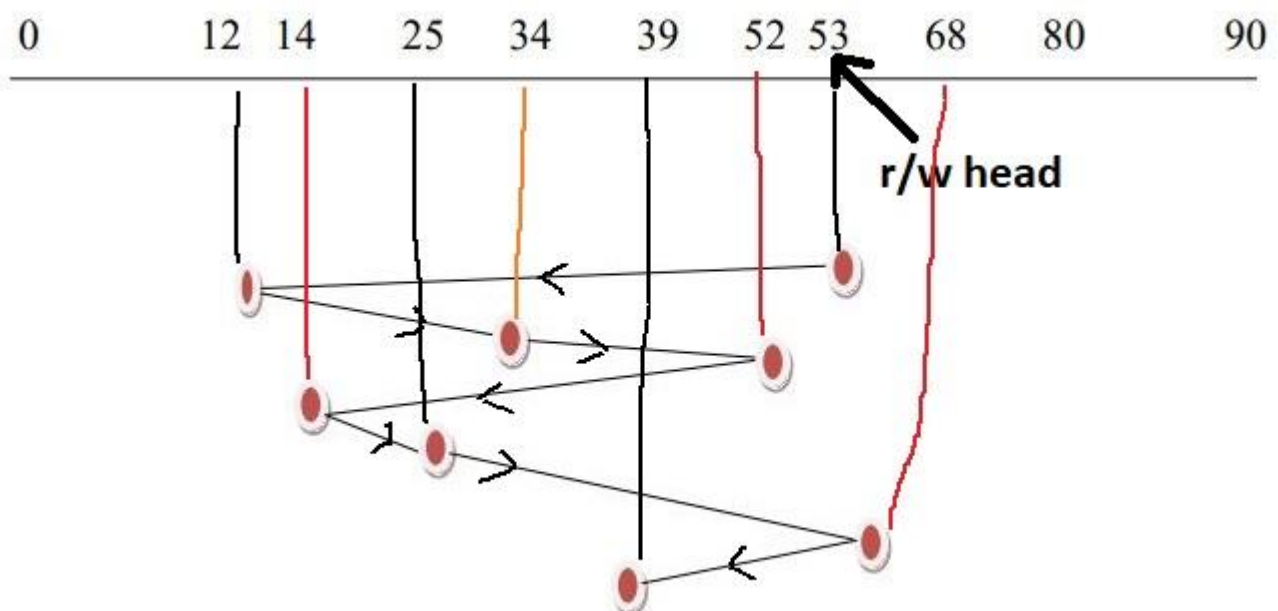
R/W head is starting at 53.

Find the number of movements using First Come First Serve scheduling.

Answer → No. of Move by the head:-

→ $(53-12) + (34-12) + (52-34) + (52-14) + (25-14) + (68-25) + (68-39)$

→ 112 (move)



Q2. A disk queue with requests for I/O to blocks on cylinders 23, 89, 132, 42, 187 With disk head initially at 100. Find the number of movements using First Come First Serve scheduling.

- a. 415 b. 421 c. 420 d. 425 e. 430

Q3. Queue = 98,183,37,122,14,124,65,67

Head starts at 53. Total moves = ?

- a. 610 b. 620 c. 640 d. 630 e. 650

2. SSTF(Shortest Seek Time First)

- Selects the request with the minimum seek time from the current head position.
- SSTF scheduling is a form of SJF scheduling; may cause starvation of some requests.

✓ Advantage:

- Very Efficient in seek move
- Less average response time and waiting time.
- Increase Throughput.

✓ Disadvantage :

- Overhead to find out the closet requirement

Q.Consider the following disk request sequence for a disk with 90 tracks 12, 34, 52, 14,25,68,39 R/W head is starting at 53.

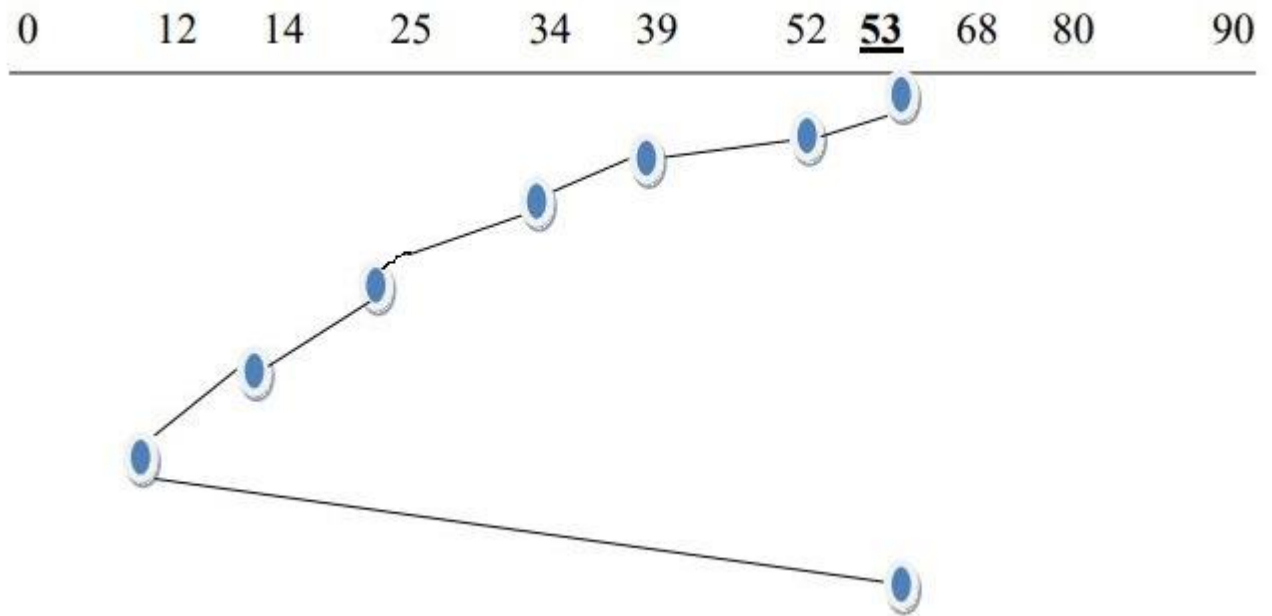
Find the number of movements using SSTF scheduling.

Answer →

No. of Move by the head:-

→ $(53-52)+(52-39)+(39-34)+(34-25) + (25-14)+(14-12)+(68-12)$

➔ 97 (moves)



Q4. Queue = 98,183,37,122,14,124,65,67

Head starts at 53

- a. 232 b. 233 c. 234 d. 236 e. 237

Q5. Queue = 23, 89, 132, 42, 187 , Head starts at 100

- a. 271 b. 272 c. 269 d. 274 e. 273

❖ Elevator Algorithms

- Algorithms based on the common elevator principle.
(इस algorithm में head को प्रारंभ से लेकर अंत तक,disk को पूरा scan करना होता है |)
- Four combinations of Elevator algorithms
- Service in both directions or in only one direction.
- Go until last cylinder or until last I/O request.

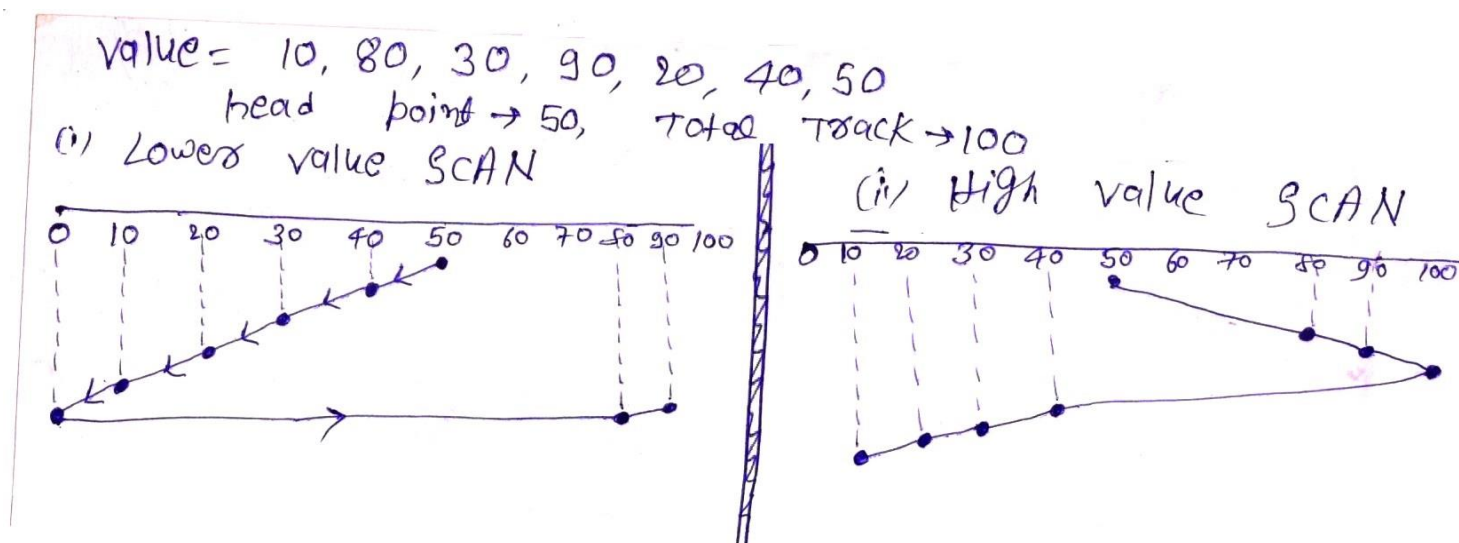
Go until Direction	Go until the last cylinder	Go until the last request
Service both directions	Scan	Look
Service in only one direction	C-Scan	C-Look

3. SCAN

- Head start at one end of the disk & moves towards the other end.

(scan में head जहाँ हैं वहाँ से यदि Direction low value हैं तो ट्रैक 0 तक जायेगा पर यदि डायरेक्शन high हैं तो वह high point तक जायेगा)

ध्यान रखें उदाहरण से -)



यहाँ 100 तक scan नहीं हुआ | और यहाँ 0 तक scan नहीं हुआ |

- Some time is called elevator Algorithm.
- The direction of the head is reversed and process continues.

✓ Advantage

- Low variance and average waiting time.
- Simple easy to understand.

✓ Disadvantage

- Long waiting time for location just visited by head.

Q. Consider the following disk request sequence for a disk with 90 tracks 12, 34, 52, 14, 25, 68, 39 R/W head is starting at 53 heigher value/Large value.

Answer →

= No. of Move by the head

$$= (68-53) + (90-68) + (90-52) + (52-39) + (39-34) + (34-25) + (25-14) + (14-12)$$

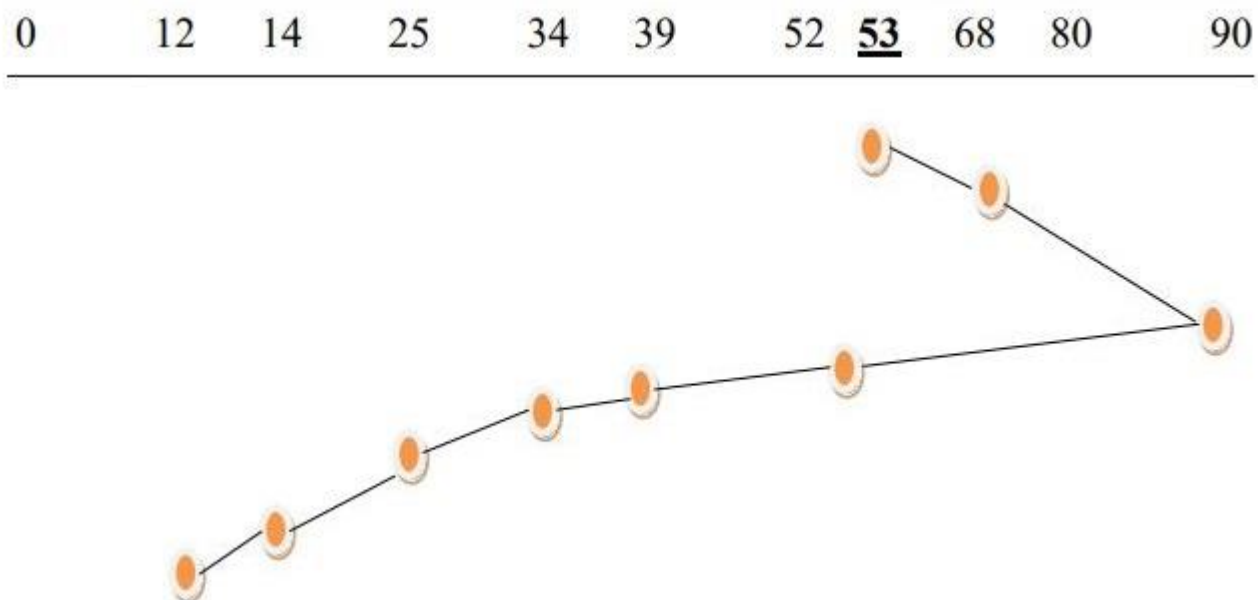
$$= 115(\text{MOVES}).$$

OR

$$= (\text{Total track value} - \text{head start value}) + (\text{Total track value} - \text{low value})$$

$$= (90-53) + (90-12)$$

$$= 115$$



Q. In this above question if R/W head time takes 1 nano second(ns) to move from one track to another then total time taken.....? Ans→ 115nm

Q6.Queue = 23, 89, 132, 42, 187 , Head starts at 100 direction lower value.

- a. 287 b. 288 c. 289 d. 286 e. 285

Q7. Queue = 98,183,37,122,14,124,65,67 Head starts at 53 direction lower value.

- a. 207 b. 208 c. 209 d. 210 e. 206

4. C-SCAN

- Head start at one end of the disk & moves towards the other end. (इसमें दोनों तरफ last end तक जाना है)

- The direction of the head is reversed and Head reaches First end without satisfy any request.

✓ Advantage:

- Provide uniform waiting time.
- Better Response Time.

✓ Disadvantage :

- More Risk ,compare to simple Scan Algorithm

Q. Consider the following disk request sequence for a disk with 90 tracks 12, 34, 52, 14,25,68,39 R/W head is starting at 53 direction larger value.

Answer →

= No. of Move by the head:

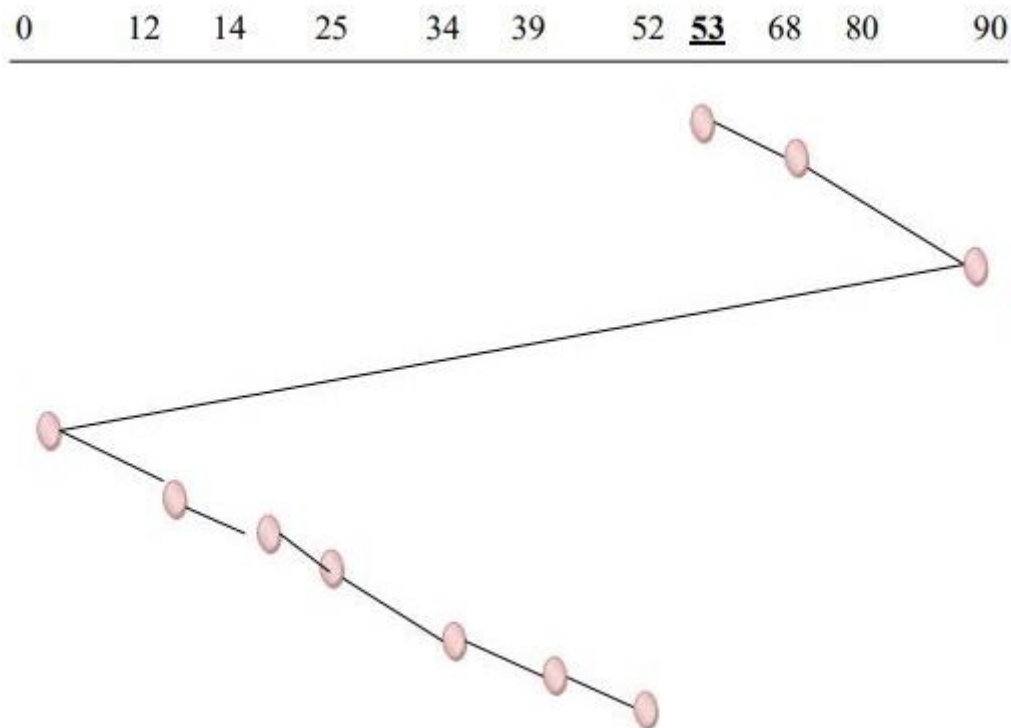
$$= (68-53)+(90-68)+(90-0)+(12-0)+ (14-12)+(25-14)+(34-25)+(39-34) \\ + (52-39)$$

$$= 179(\text{MOVES}).$$

OR

$$= (90 - 53) + (90 - 0) + (52 - 0)$$

$$= 179$$



Q8. Queue = 23, 89, 132, 42, 187 , Head starts at 100 direction lower value.

- a. 364 b. 365 c. 366 d. 367 e. 368

Q9. Queue = 82,170,43,140,24,16,190 Head starts at 50 direction large value.

- a. 390 b. 395 c. 392 d. 391 e. None

5. LOOK

- LOOK It is same as scan algorithm but instead of going till last track we go till last request & then change direction.

(look में head end तक नहीं जायेगा ना lower value में न high value में | head वहाँ तक ही जायेगा जहाँ तक request किया गया हो)

Scan	Look
इसमें lower value में head low end तक जायेगा And high value में head high end तक जायेगा	इसमें lower value में head low end तक नहीं जायेगा, जहाँ तक request हुआ है वहाँ तक ही जायेगा

✓ **Advantage :-** Better performance compare to scan algorithm

✓ **Disadvantage :-** Overhead to find the last request

Q. Consider the following disk request sequence for a disk with 90 tracks 12, 34, 52, 14,25,68,39 R/W head is starting at 53 direction high value.

Answer:

= No. of Move by the head:

= $(68-53)+(68-52)+(52-39)+(39-34) + (34-25)+(25-14)+(14-12)$

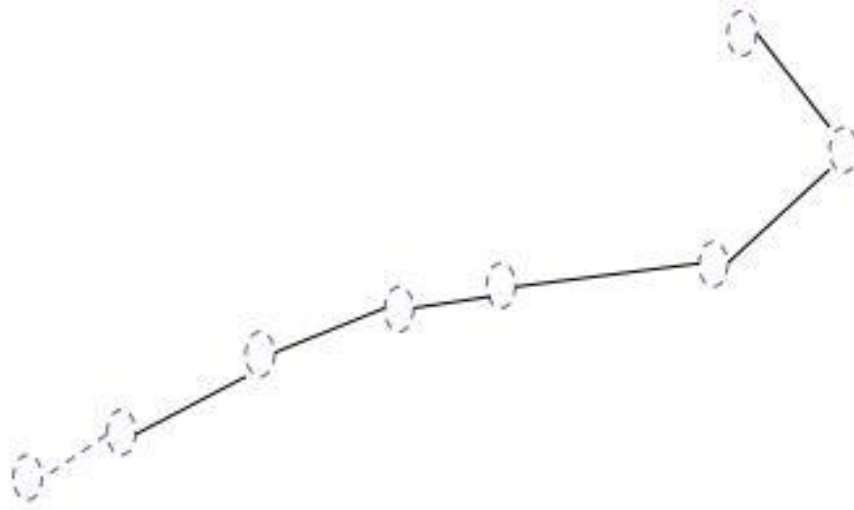
=71(MOVES).

OR

= $(68-53) + (68 - 12)$

= 71

0 12 14 25 34 39 52 53 68 80 90



Q10. Queue = 23, 89, 132, 42, 187 , Head starts at 100 direction lower value.

- a. 243 b. 240 c. 242 d. 239 e. 241

Q11. Queue = 82,170,43,140,24,16,190 Head starts at 50 direction large value.

- a. 314 b. 315 c. 313 d. 312 e. 316

6. C-LOOK

- It takes the advantage of both C-Scan and Look algorithm.
- We will satisfy request only in are direction.
- We will go till last request & return but not till last track.
- ✓ **Advantage:** More uniform waiting time compare to LOOK algorithm.
- ✓ **Disadvantage :** More Overhead in calculate.

Q. Consider the following disk request sequence for a disk with 90 tracks 12, 34, 52, 14,25,68,39 R/W head is starting at 53 direction high value.

Answer:

= No. of Move by the head:

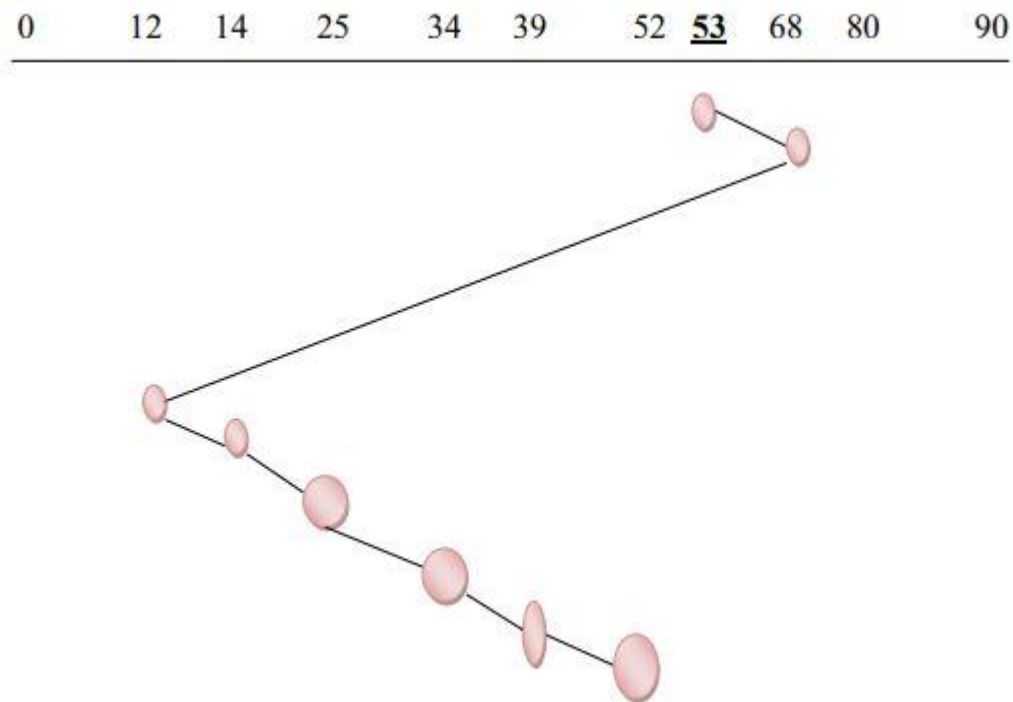
$$= (68-53)+(68-12)+(14-12)+(25-14) + (34-25)+(39-34)+(52-39)$$

$$=111(\text{MOVES}).$$

OR

$$= (68-53) + (68 - 12) + (52 - 12)$$

$$= 111$$



Q12. Queue = 82,170,43,140,24,16,190 Head starts at 50 direction large value.

- a. 340 b. 341 c. 342 d. 343 e. 339

Practical online- <https://www.binaryvisual.com/operating-systems/disk-scheduling>

Algorithm Comparison

Algorithm	Seek Time	Fairness	Starvation	Complexity
FCFS	Poor	Excellent	None	Low
SSTF	Good	Poor	Possible	Medium
SCAN	Good	Good	None	Medium
C-SCAN	Fair	Excellent	None	Medium
LOOK	Excellent	Good	None	Medium
C-LOOK	Excellent	Excellent	None	High